

Yiran (Kevin) Zhou

(347) 200-0063 | yrzhou@umich.edu | [kevin-yiran-zhou.github.io](https://github.com/kevin-yiran-zhou) | [linkedin.com/in/kevin-yiran-zhou](https://www.linkedin.com/in/kevin-yiran-zhou)

EDUCATION

University of Michigan | Ann Arbor, MI

Ph.D. in Robotics

Jan 2026 - Present

M.S. in Robotics

Aug 2024 - May 2026

- Research Advisor: James Weiland
- Research Interests: Assistive Technology, HCI (Human-Computer Interaction)

New York University | New York, NY

Sep 2020 - May 2024

B.S. in Mathematics, minor in Computer Science

GPA: 3.83/4.0

- Honors: Cum Laude, School of Engineering Dean's List

RESEARCH EXPERIENCE

AVA: A smartglass-integrated assistive system for people with visual impairment

Sep 2024 - Present

BioElectronic Vision Lab | Ann Arbor, MI

- Developed an on-device multimodal assistive Android application integrating speech, computer vision, and haptic feedback to support navigation and spatial awareness for visually impaired users via custom smartglasses.
- Implemented an on-device low-latency voice interaction pipeline using **Picovoice Porcupine** (wake-word detection), **Google Speech-to-Text**, and lightweight **Small Language Models**, enabling hands-free control and intent-driven function triggering.
- Built real-time perception modules including monocular depth-based obstacle detection (**Depth Anything**) and object detection for door handles (**YOLOv8**), optimized for mobile deployment.
- Designed adaptive feedback strategies combining audio and vibration cues to guide users in obstacle avoidance and target localization under varying environmental conditions.
- Conducted user studies with 10 visually impaired participants, iterating on system design based on usability feedback and improving task success rate and user experience.

UNav: A vision-based navigation system for the visually impaired

Sep 2022 - May 2024

VMIL/REACTIV Lab | New York, NY

- Engineered and deployed a navigation mode selector and an alert system in the UNav Android app, significantly improving user experience in trials with 12 visually impaired participants. Participated in a demonstration to the NSF, helped the project team secure a grant.
- Collected 360° map videos from 3 school buildings, 3 hospital sites, and 2 residential houses to create a system database. Processed videos and built map databases using **OpenVSLAM** and **PyTorch**.
- Debugged and improved the UNav Android app and Jetson Nano wearable system through extensive real-world testing, identifying software bugs and correcting map data to enhance navigation stability and user safety.
- Enhanced and maintained Python programs and shell scripts on a **Linux** server and the NYU HPC clusters.

WORK EXPERIENCE

New York University, Embodied AI Team | New York, NY

Sep 2023 - Dec 2023

Software Developer

- Employed ROS and Python to craft robot programs, encompassing robot control games and navigation systems.
- Orchestrated the deployment of LiDAR-based SLAM using ROS and Gazebo to generate detailed and accurate navigational maps in diverse environments, enhancing the precision of robotic navigation.
- Integrated Large Language Models (**LLMs**) to refine robot movement controls, significantly improving response accuracy and operational efficiency in dynamic settings.

SKILLS

- **Programming Skills:** Python (Numpy, Pandas, Pytorch, Tensorflow, Scikit-learn, ROS), C++, Java (Android Studio), Linux, Git, MySQL, Verilog, MATLAB, LaTeX.
- **Languages:** English (Fluent), Mandarin (Native).
- **Interests:** Photography, travelling, marathon running.